EXPERIMENTS IN MODERN PHYSICS

Second Edition

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If the outer layers of the source are much cooler than the middle ones, the width of the particular energy level (due to the Doppler effect) is smaller in the outer layers and absorption takes place only at the central frequency with almost none in the wings. The result is a "self-reversed" line as shown in Fig. 6.13b. This effect is very pronounced in the sodium D lines, and when it is viewed with a high-resolution instrument, the line exhibits a doublet structure that is frequently mistaken for hyperfine structure. One can verify the origin of the effect because it varies with the voltage used to excite the source.

6.5. THE ZEEMAN EFFECT OF THE GREEN LINE OF ^{198}Hg

6.5.1. Equipment and Alignment

We now discuss the observation of the Zeeman effect on the $\lambda = 546.1$-nm line of ^{198}Hg. The choice of the green line is due to its predominance in the mercury spectrum, and the ease with which it can be observed. In an external magnetic field, it is split into nine components, as discussed in detail in Section 6.2.2. In the present observations, a polarizer parallel to the magnetic field was used, so that only three of the nine components (the π light) appeared. Furthermore, natural mercury exhibits in the green line a large number of hyperfine structure components, and each of them forms a Zeeman pattern. To avoid a multiplicity of components in one spectral line, a separated isotope of mercury was used as the source. ^{198}Hg is well suited for this purpose since $I = 0$, and therefore it exhibits no hyperfine structure.

The optical system used for this investigation is shown in Fig. 6.14. The Fabry–Perot was crossed in the parallel-beam method with a small constant-deviation spectrograph (see Chapter 1). The etalon and lenses are all mounted on an optical bench to which the spectrograph is rigidly attached. The pair of lenses L_1 forms the light from the source into a parallel beam, while the pair L_2 focuses the Fabry–Perot ring pattern onto the spectrograph slit; the effective focal length of L_2 is 8 cm, and a further magnification of 2 takes place in the spectrograph.

The discharge tube is mounted vertically, as is the spectrograph slit; the slit width was 1 mm. It is clear that in this arrangement not only is the ring pattern focused onto the spectrometer slit but also the image of the source. A sheet of Polaroid film that could be rotated at will was used as a polarizer.

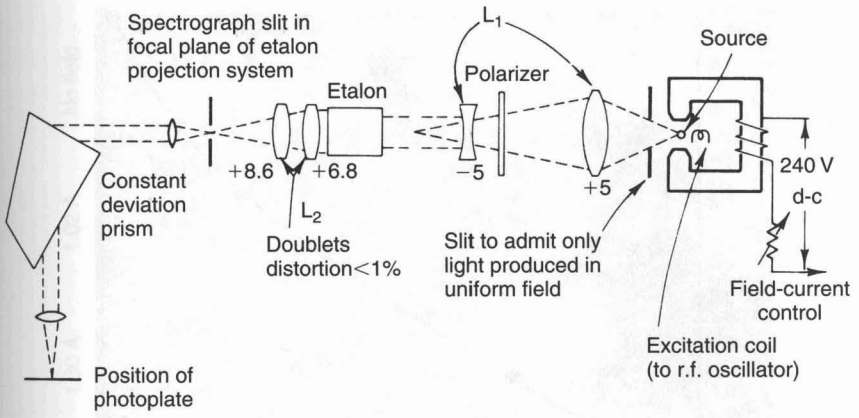


FIGURE 6.14 Experimental arrangement used for observing the Zeeman effect with a Fabry–Perot etalon, crossed by a constant-deviation prism spectrograph.

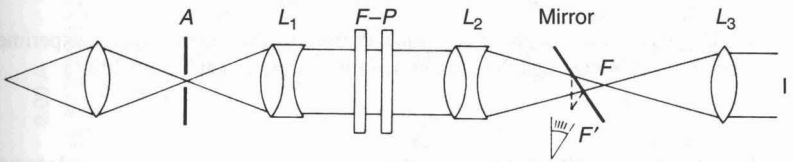


FIGURE 6.15 Optical arrangement for aligning a Fabry–Perot etalon. Rough adjustment is made by viewing the image formed by L_2. Final adjustment is made by viewing the etalon from the point F (or F').

The spacing of the Fabry–Perot etalon is $t = 0.5002$ cm; namely, the free spectral range is FSR = 30 GHz. It is important to adjust the plates carefully for parallelism. This can be done either by viewing through the spectrograph with a frosted glass in the focal plane, and adjusting for the best quality of the pattern, or by a much more sensitive arrangement as shown in Fig. 6.15. A very small aperture (less than 1 mm in diameter) is placed at the position of the source and illuminated with an intense sodium lamp. The Fabry–Perot plates are adjusted to be normal to the optical axis by bringing the image of A reflected by the etalon back onto A. Next, L_3 is adjusted until a series of multiple images of A appears when the observer is located at I; the plates of the etalon can then be roughly adjusted for parallelism by bringing all the images into coincidence. The final adjustment is made by removing L_3 so that the observer locates his eye at F (or a mirror can be used); then fringes of equal width do appear

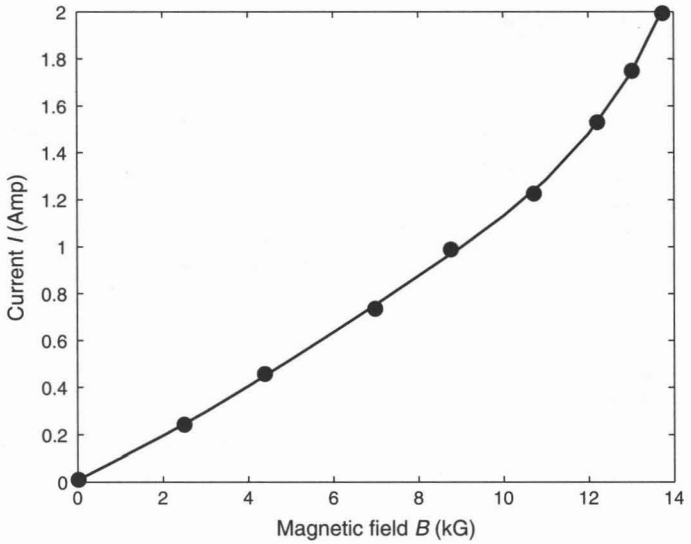


FIGURE 6.16 Calibration of the electromagnet used in the Zeeman effect experiment. The magnetic field is plotted against current; note the saturation at high fields.

parallel to the base of the wedge formed by the two plates. As the plates are moved into parallelism, the fringes become broader and finally the whole image of the aperture A seems to have a uniform illumination (bright or dark depending on the exact value of $n_0 = 2t/\lambda$). It is equally important that the ring pattern be in sharp focus at the plane of the photographic plate. For this experiment Kodak Royal-Pan film was used.

The electrodeless discharge tube was placed in a magnetic field. A small iron core electromagnet powered by a 220-V DC supply was used to produce the field. The diameter of the pole faces was only $1\frac{1}{2}$ in., and a small gap ($\frac{1}{2}$ in.) was used. By tapering the pole faces, higher magnetic fields can be achieved but this reduces the effective area of the field as well as the homogeneity. The magnetic field was measured with a "flip coil" and the calibration of field against current is given in Fig. 6.16. It is seen that field strengths of 1.2 T could be reached.

6.5.2. Data on the Zeeman Effect

The data presented below were obtained by students. Figure 6.17 shows the 546.1-nm Hg line photographed at various magnet settings. As explained

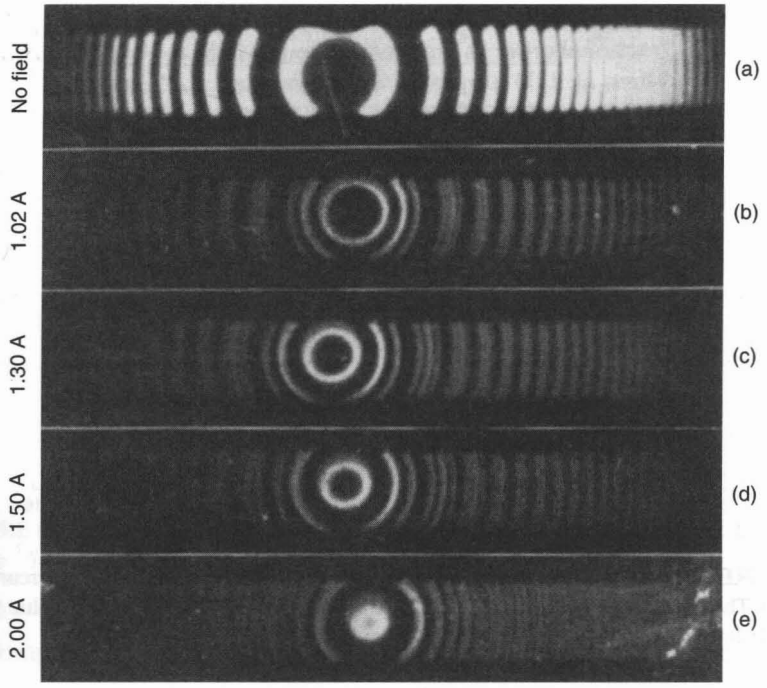


FIGURE 6.17 Fabry–Perot patterns showing the Zeeman effect of the green line of mercury. (See the text for additional details.) (a) No magnetic field applied. (b–e) A magnetic field of progressively greater strength is applied. Note the splitting of the original line into a triplet of increasing separation.

earlier, the source contains a single isotope, and the polarizer allows only the observation of π light. We note that the fringes are rather broad, but it can clearly be seen that when the field is applied the single–line pattern breaks up into a triplet, the separation between the components of the triplet becoming larger with increasing field.

The initial step in the reduction of the data is the measurement of the diameters (or radii) of the rings. To this effect a traveling microscope was used, and readings were taken directly off the plate; care must be taken to ensure that the travel of the microscope is indeed along the diameter of the rings and that the crosshairs are properly oriented. When the fringes in the pattern are as broad as those in Fig. 6.17, it is much more accurate to measure the two edges and take the average rather than try to set the crosshairs in the center of the fringe. The ring radii squared in the absence of the field provide the calibration of the data.

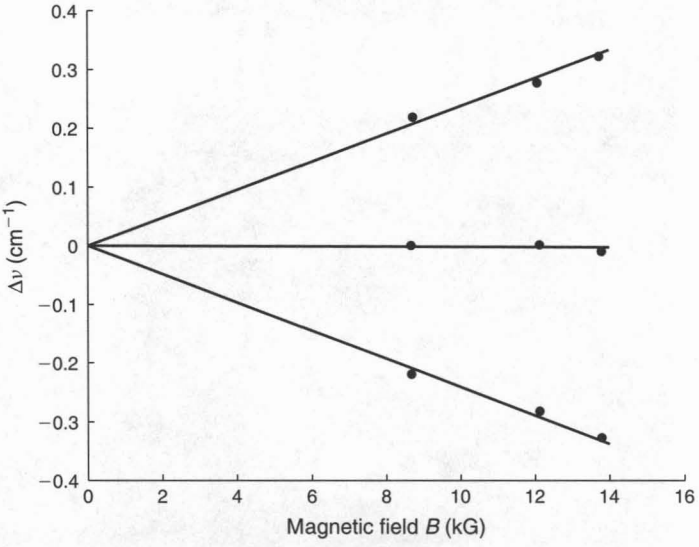


FIGURE 6.18 Results obtained on the Zeeman effect of the green line of mercury (see text). The observed displacement of the three components from the zero field value (of the single line) is plotted against the magnetic field.

Next the radii of the rings for the exposures taken at 1.0, 1.5, and 2.0 A were analyzed, and it was found that the central line is not shifted. However, the following shifts are observed for the outer rings for the 1.0-A data:

$$\Delta\nu_+ = 6.81 \text{ GHz} \qquad \Delta\nu_- = 6.60 \text{ GHz}.$$

The complete set of data is plotted in Fig. 6.18, and we see that as predicted the spacing varies linearly with the field, yielding

$$\Delta\nu = (7.2 \text{ GHz}) \times B(\text{T}). \tag{6.30}$$

The green line of Hg (546.1 nm) connects the $^3S_1$ state to the $^3P_2$. Its Zeeman splitting is shown in Fig. 6.19 where the g factors have been calculated according to Eq. (6.17). Since the polarizer was set to select only components arising in transitions with $\Delta m = 0$, we expect to observe only the three central components, which will be separated by

$$\Delta\nu = \frac{\mu_B}{h}(g_i - g_f)B = \frac{1}{2}\frac{\mu_B}{h}B. \tag{6.31}$$

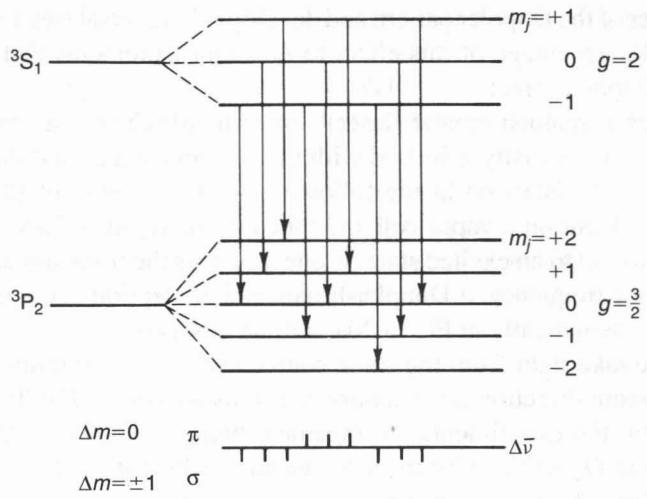


FIGURE 6.19 The Zeeman multiplet splitting of the 546.1-nm green line of Hg. It arises from a $^3S_1$ to $^3P_1$ transition.

By comparing with the experimental result of Eq. (6.30), we obtain

$$\mu_B = 5.95 \times 10^{-11} \text{ MeV/T}$$

in good agreement with the accepted value of

$$\mu_B = 5.79 \times 10^{-11} \text{ MeV/T}.$$

From these data we conclude that indeed spectral lines are split into components when the source is placed in a magnetic field. Further, the splitting observed was in excellent agreement with the theory of the anomalous Zeeman effect; the normal Zeeman effect can be excluded, since the energy difference between the components of the line was not $\mu_B B$ but $\frac{1}{2}\mu_B B$; compare to Eq. (6.1).

6.6. SATURATION ABSORPTION SPECTROSCOPY OF RUBIDIUM

6.6.1. Introduction

We mentioned in Section 6.4 that if an intense spectral line is passed through a region of dense atomic vapor of the same element it may become absorbed